The University of Lahore

Department of Computer Science& IT

Midterm Examination

Summer-2023

Course Title:	Object Oriented Programming	Course Code:	CS02203||11	Credit Hours:	4
Course Instructors:	Ms. Mariya Bibi	Program Name:	BSCS		
Time Allowed:	**70 Minutes**	**Maximum Marks:**	**30**		
Date:	**7th Aug, 2023**	Course Mentor:	Dr. Muhammad Atif		
Student's Name:		Signature:			
Reg. No:		Section:			

Important Instructions / Guidelines:

- Attempt the objective part on the question paper before moving to the subjective part.
- Solve the subjective part on the answer book provided separately.

[Time Allowed: 25 minutes]

Objective Part

[Marks = 14]

Question # 1: Identify the errors and write the output for the following **[Marks = 3+3+4+4]**

Code	**Errors**	**Output (if any)**

A:
class Student{
  private:
    static int n;
  public:
  Student(){
    n++;}
  void creation(){
 cout<<"you Have Created "<<n<<" objects "<<endl;
  return n;}
};
int main(){
    Student s1,s2;
    return 0; }		

B:
class equal
{
int one, two;
public:
bool equal();
print();
BB(int, int);
}		

C:
#include<iostream>
using namespace std;
class dummyClass {
private:
int n;
public:
dummyClass() {
cout<<"\n I am Default constructor"; }
dummyClass(int x) {
cout<<"\n I am dummy constructor";
n=x; }
int getter() {
return n; }
int main(){
int z;
cin>>z;
dummyClass dc1, dc2(z);
dc.getter();
dc.dummyClass();
dc1.getter()
return 0; }		

D:
class c1{
public:
c1() {
cout<<"this is c1 constructor"<<endl; }
void function(){
cout<<"I am in c1"<<endl; }
};
class c2: public c1{
public
c2() {
cout<<"this is c2 constructor" <<endl; }
void function(){
cout<<"I am in c2"<<endl; }
};
class c3: public c1 {
public:
void function(){
cout<<"I am in c3"<<endl; }
};
int main(){
c2 test;
c3 test2;
test2.function;
test.function();
test2.c2::function()
return 0;
}		

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Reg. No:		Section:			
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- Solve the subjective part on the answer sheet provided separately.

Subjective Part [Total Marks: 16]

Question # 2 Write the C++ code of the following statement. [8+8Marks]

A. Write a C++ program that contains a class named "Mother," which has two properties: "eye_color" and "skin_color." By default, the eye color is brown, and the skin color is whitish. Another class named "Father" is defined, which includes two properties: "eye_color" and "skin_color," along with a parameterized function father_height() that takes a value from the user during function call. By default, the properties of the Father class are gray eye color and sand skin color.

Next, create a derived class named "Son" that inherits the eye color and skin color from his mother and the height from his father. Implement a function named display_son_prop() to display the properties of the son. Additionally, create another derived class named "Daughter," which inherits the skin color from her mother and the eye color from her father. Implement a function named display_daughter() to display the properties of the daughter.

Finally, create objects of the "Son" and "Daughter" classes to display all properties of both children.

B. Write a C++ program that defines a class named "Employee." The class should have three data members: "empName" for the employee's name, "empDesignation" for the employee's designation, and "empSalary" for the employee's salary. Create a parameterized constructor to initialize all three data members, with values provided by the user. Additionally, write getter() and setter() functions for all three data members in the Employee class.

Implement a member function in the class named increment() to calculate the incremented salary and the total salary after increment for an employee. In this function, the user will enter the percentage of increment in salary through the keyboard, and it will display the total salary after the increment.

In the main function, create an object for the Employee class and call the increment() function along with the getter functions for name, designation, and salary.